

FIN305 Risk Management for Business

Week 1 Questions

Practice questions and problems

The numbers in parentheses after the questions correspond to the original numbering used in the textbook, which is intended to assist students in easily finding the relevant questions within the book.

1. An investment has probabilities 0.1, 0.2, 0.35, 0.25, and 0.1 of giving returns equal to 40%, 30%, 15%, -5%, and -15%. What are the expected returns and the standard deviations of returns? (1.1)
2. What is the difference between systematic and nonsystematic risk? Which is more important to an equity investor? Which can lead to the bankruptcy of a corporation? (1.4)
3. The return from the market last year was 10% and the risk-free rate was 5%. A hedge fund manager with a beta of 0.6 has an alpha of 4%. What return did the hedge fund manager earn? (1.14)
4. A portfolio manager has maintained an actively managed portfolio with a beta of 0.2. During the last year, the risk-free rate was 5% and major equity indices performed very badly, providing returns of about -30%. The portfolio manager produced a return of -10% and claims that in the circumstances it was good. Discuss this claim. (1.18)
5. What is the difference between entering into a long forward contract when the forward price is \$50 and taking a long position in a call option with a strike price of \$50? (5.3)

6. An investor enters into a short forward contract to sell 100,000 British pounds for U.S. dollars at an exchange rate of 1.3000 U.S. dollars per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is (a) 1.2900 and (b) 1.3200? (5.5)
7. The current price of a stock is \$94, and three-month European call options with a strike price of \$95 currently sell for \$4.70. An investor who feels that the price of the stock will increase is trying to decide between buying 100 shares and buying 2,000 call options (= 20 contracts). Both strategies involve an investment of \$9,400. What advice would you give? How high does the stock price have to rise for the option strategy to be more profitable? (5.33)

Solutions

Each week, students should attempt the provided exercise questions after lectures. Solutions can be found at the end of the textbook to check your answers. For those questions to which no solutions can be found in the textbook, the detailed solutions/explanations will be shown in the present file as follows. Selected questions will be discussed in tutorials, where the instructor will offer more detailed solutions. Recordings of the related tutorials will be made available to all students via Learning Mall.

Question (4)

- When the expected return on the market is -30%, the expected return on a portfolio with a beta of 0.2 is $0.05 + 0.2 \times (-0.30 - 0.05) = -0.02$, or -2%. The actual return of -10% is worse than the expected return. The portfolio manager has achieved an alpha of -8%!

Question (7)

- The investment in call options entails higher risks but can lead to higher returns. If the stock price stays at \$94, an investor who buys call options loses \$9,400 whereas an investor who buys shares neither gains nor loses anything. If the stock price rises to \$120, the investor who buys call options gains $2000 \times (120 - 95) - 9400 = 40,600$. An investor who buys shares gains $100 \times (120 - 94) = 2600$. The strategies are equally profitable if the stock price rises to a level, S , where $100 \times (S - 94) = 2000 \times (S - 95) - 9400$, which leads to $S = 100$. The option strategy is therefore more profitable if the stock price rises above \$100.